

Femtomodels for EEG Artifact Removal

A Parameter Lower-Bound for Generalisable EOG Denoising

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Motivation

The Problem: EOG Artifacts in EEG

Ocular artifacts are a principal source of non-neural contamination in scalp EEG:

- Spread broadly across electrodes
- Dominate lowest frequency bands
- Distort power, waveforms, and connectivity [1, 2]

Classical solutions (filtering, regression, ICA) trade off neural fidelity or require manual tuning.

Deep learning denoisers achieve strong results but use **100k–9M+ parameters**.

Key Question

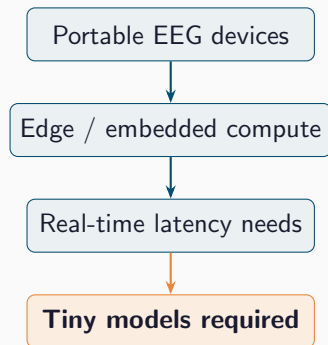
How many parameters do we *actually need* for effective EOG removal?

Our Answer

~**2k parameters** suffice.

Beyond this, gains are marginal.

Why Capacity Matters



Existing denoisers:

Architecture	Params
IC-U-Net	~280k
UNet++ variants	~500k
MultiResUNet3+	~1.3M
Transformer-based	2–9M
FemtoEOGClean	2.5k

100×–1000× smaller than typical designs.

Theory

Signal Model & Optimal Denoiser

Signal model: Multichannel EEG $x \in \mathbb{R}^{C \times T}$:

$$x = s + Bu, \quad u \in \mathbb{R}^{r \times T}, \quad B \in \mathbb{R}^{C \times r}$$

where s is clean EEG, u contains r latent ocular sources, and B is the spatial mixing matrix.

Wiener filter (MSE-optimal linear estimator):

$$H^*(\omega) = S_{ss} (S_{ss} + B S_{uu} B^\top)^{-1}$$

By Woodbury:

$$H^*(\omega) = I - \underbrace{B(S_{uu}^{-1} + B^\top S_{ss}^{-1} B)^{-1} B^\top S_{ss}^{-1}}_{\text{rank-}r \text{ correction}}$$

$\Rightarrow$ The optimal denoiser is **identity minus a rank- r operator**, applied only on the EOG band Ω_{EOG} .^[6, 7]

Low Degrees-of-Freedom Realisation

Constructive form:

$$\mathcal{T}_\theta(x) = x - U(h * (U^\top x))$$

with $U \in \mathbb{R}^{C \times r}$ (spatial demixer) and $h = \{h_k\}_{k=0}^{L-1}$ an $r \times r$ FIR filter.

Degrees of Freedom

$$\text{DoF} \leq \underbrace{rC}_{\text{spatial}} + \underbrace{r^2L}_{\text{temporal}}$$

- Spatial demixer: rC coefficients
- Temporal FIR: r^2L coefficients

Approximation Guarantee

For any $\varepsilon > 0$, $\exists L, h$ such that:

$$\sup_{\omega} \|H^*(\omega) - H_{\mathcal{T}_\theta}(\omega)\|_2 \leq \varepsilon$$

with $L = O(1/\Delta f)$.

$\Rightarrow$ A small number of free parameters can approximate the Wiener map arbitrarily well.

Capacity Bound & Plateau Law

CNN saturation point. Accounting for architectural overhead $\kappa \geq 1$:

$$P_{\text{sat}} = \kappa (rC + r^2L)$$

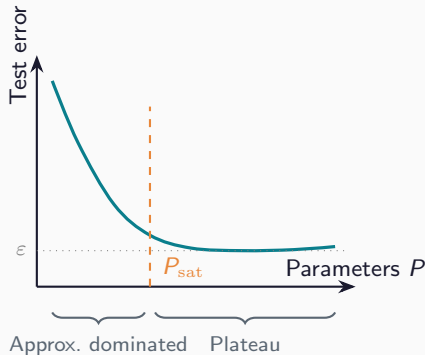
Plateau Law (Theorem 1)

If $P \geq P_{\text{sat}}$: $\mathcal{E}_{\text{test}}(P) \leq \varepsilon + \tilde{O}(\sqrt{P/n})$

- $P < P_{\text{sat}}$: **approx. error** dominates
- $P > P_{\text{sat}}$: **estim. error** dominates $\rightarrow$ **plateau**

With $r \in \{1, 2\}$, moderate C, L :

$P_{\text{sat}} \approx$ **low thousands** $\rightarrow$ empirical elbow at $\sim 2k$.



Experiments

Femtomodel Family

Architecture: Lightweight convolutional autoencoders.

Each model defined by $\{ch, s, g\}$:

- ch : base channel count
- s : feature scaling in deeper layers
- g : grouping factor

Sweep:

- Depths: conv3 to conv6
- $ch \in \{1, 2, 4, 8, 16, 24\}$
- $s \in \{1, 2, 4, 8\}$, $g \in \{1, 2, 4, 8\}$
- **143 models** total, from 41 to ~66k

Datasets:

Training & In-domain [8]

EEGDenoiseNet (EOG subset)
30k train / 4k test pairs
2s segments, 256 Hz
SNR: -7 to $+2$ dB

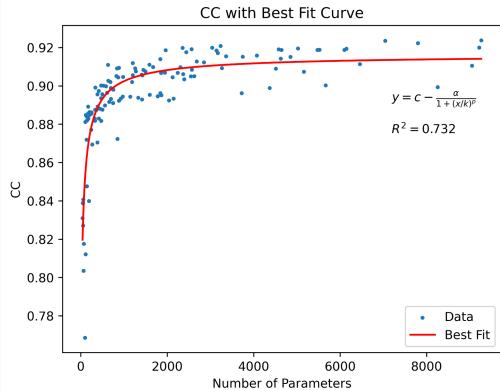
Generalisation (novel) [9]

BCI Competition IV 2a
~5k evaluation pairs
EOG-based artifact extraction
SNR: -3 to $+3$ dB

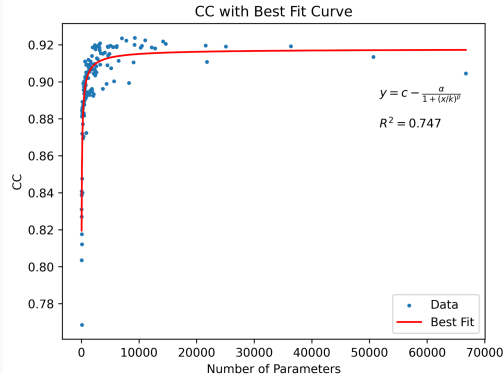
Metrics: T-RRMSE, RMSE, CC, SDR

Results

Capacity–Performance Plateau: Correlation Coefficient



(a) CC vs. parameters, <10k models

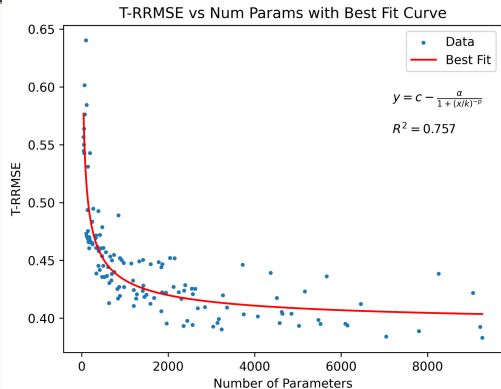


(b) CC vs. parameters, full sweep

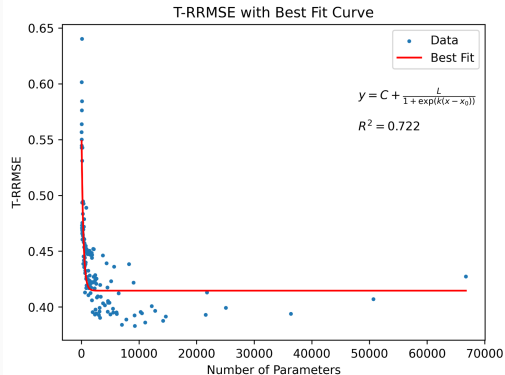
Both scales well fit by a **hill function** ($R^2 > 0.73$).

Performance saturates beyond $\sim 2k$ parameters — consistent with the theoretical P_{sat} .

Capacity–Performance Plateau: T-RRMSE



(c) T-RRMSE vs. parameters, <10k models



(d) T-RRMSE vs. parameters, full sweep

Sub-10k: hill function fit. Full range: **sigmoid** ($R^2 > 0.72$).

Elbow consistently at a few thousand parameters across all metrics.

FemtoEOGClean vs. Large Models

Model	Params	EEGDenoiseNet (in-domain)				BCI-IV 2a (generalisation)			
		T-RRMSE↓	RMSE↓	CC↑	SDR↑	T-RRMSE↓	RMSE↓	CC↑	SDR↑
6conv, $g=24$	66.7k	0.427	0.238	0.905	7.38	0.744	0.744	0.670	2.57
6conv, $g=1$	50.7k	0.407	0.227	0.914	7.81	0.722	0.722	0.695	2.83
6conv, $g=2$	36.4k	0.394	0.219	0.919	8.09	0.721	0.721	0.697	2.84
FEOGC	2.5k	0.394	0.220	0.919	8.09	0.685	0.685	0.730	3.29

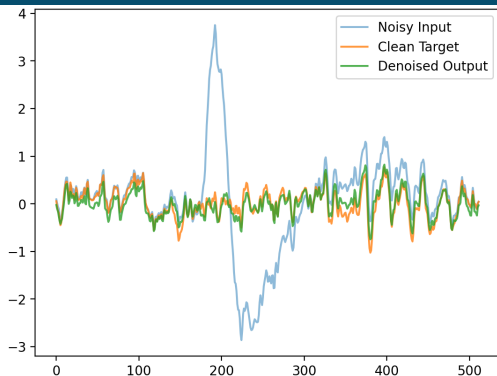
In-domain

Matches 36k model ($< 10^{-3}$ differences),
surpasses both 50k and 66k models.

Generalisation

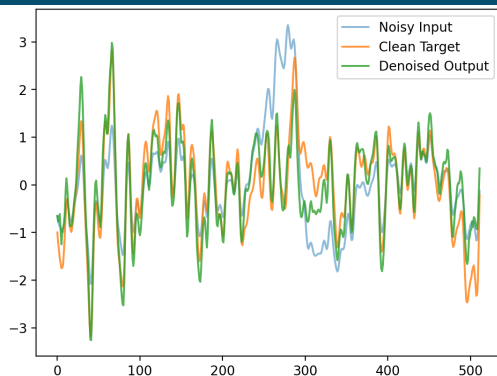
Significantly outperforms all large
models on the unseen BCI-IV dataset.

Qualitative Denoising Examples



EEGDenoiseNet holdout (629 params)

Best sub-1k model on in-domain data

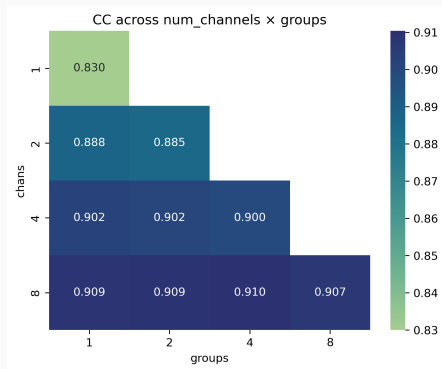


BCI-IV 2a generalisation (881 params)

Best sub-1k model on unseen dataset

Even **sub-1k parameter** models produce clean reconstructions,
both in-domain and under dataset shift.

Architectural Factors



Average CC across channel width and grouping
(aggregated over depth and scale)

Key observations:

- Pronounced sensitivity only at **smallest capacities** ($ch=1$: $CC = 0.830$)
- Once $ch \geq 4$: $CC \geq 0.90$ regardless of grouping
- Beyond $\sim 2k$ params, dependence on (ch, g) **weakens substantially**

⇒ Plateau is **task-intrinsic** (low-rank + band-limited), not architecture-specific.

Conclusion

Summary & Takeaways

What we showed:

1. **Capacity study** across 143 convolutional denoisers (41 to 66k params)
2. **Theoretical bound**: the low spatial rank ($r \leq 2$) and band-limited structure of EOG artifacts imply P_{sat} in the low thousands
3. **Empirical plateau** at $\sim 2\text{k}$ parameters, confirmed across 4 metrics
4. **FemtoEOGClean** (2.5k params): matches/surpasses $> 36\text{k}$ models in-domain, **significantly better** on generalisation

Core Message

Accurate EEG denoising does **not** require massive overparameterisation.

Femtomodels achieve it with $100\times - 1000\times$ fewer parameters.

Implications

Edge deployment
Real-time processing
Portable neurotechnology
Size-aware design $>$
overparameterisation

References

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Thank You

Questions?

<https://github.com/jeetblahiri/femtoeogclean>